quick 2 knowledge

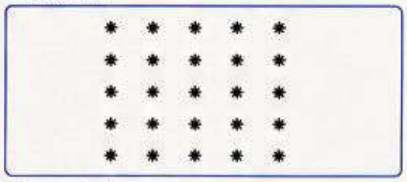
JAVA PATTERN NOTES

Java

All Important Patterns in Java

1. Solid Rectangle

Pattern:

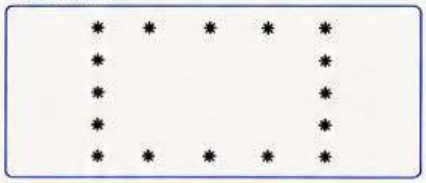


Java Code:

```
int rows = 6, cols = 5;
for (int i = 1; i <= rows; i++) {
   for (int j = 1; j <= cols; j++) {
       System.out.print("* ");
   }
   System.out.println();
}
```

2. Hollow Rectangle

Pattern:

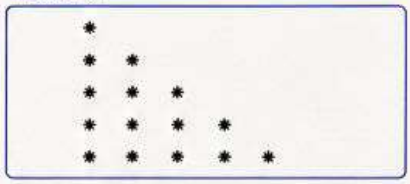


Java Code:

```
int rows = 6, cols = 5;
for (int i = 1; i <= rows; i++) {
   for (int j = 1; j <= cols; j++) {
       if (i == 1 || i == rows ||
           j == 1 || j == cols)
           System.out.print("* ");
       else
           System.out.print("  ");
   }
   System.out.println();
}
```

3. Half Pyramid

Pattern:

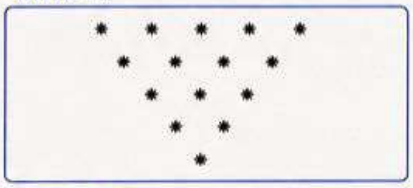


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++) {
       System.out.print("* ");
   }
   System.out.println();
}
```

4. Inverted Half Pyramid

Pattern:

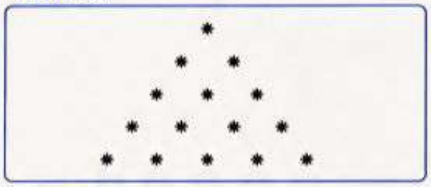


Java Code:

```
int n = 5;
for (int i = n; i >= 1; i--) {
   for (int j = 1; j <= i; j++) {
       System.out.print("* ");
   }
   System.out.println();
}
```

5. Full Pyramid

Pattern:

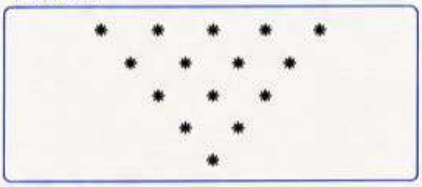


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= n - i; j++) {
       System.out.print(" ");
   }
   for (int j = 1; j <= (2 * i - 1); j++) {
       System.out.print("* ");
   }
   System.out.println();
}
```

6. Inverted Full Pyramid

Pattern:



Java Code:

```
int n = 5;
for (int i = n; i >= 1; i--) {
   for (int j = 1; j <= n - i; j++) {
       System.out.print(" ");
   }
   for (int j = 1; j <= (2 * i - 1); j++) {
       System.out.print("* ");
   }
   System.out.println();
}
```

7. Floyd's Triangle

Pattern:

```
1
2  3
4  5  6
7  8  9  10
11 12 13 14 15
```

Java Code:

```
int n = 5, num = 1;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++) {
       System.out.print(num + " ");
       num++;
   }
   System.out.println();
```

8. 0-1 Triangle

Pattern:

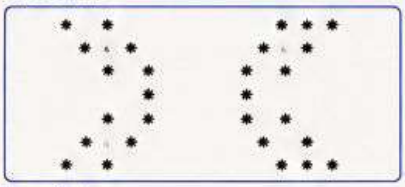
```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++) {
       if ((i + j) % 2 == 0)
           System.out.print("1 ");
       else
           System.out.print("0 ");
   }
```

9. Butterfly Pattern

Pattern:



Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++)  System.out.print("* ");
   for (int j = 1; j <= 2 * (n - i); j++) System.out.print("  ");
   for (int j = 1; j <= i; j++)  System.out.print("* ");
       System.out.println();
}
// lower part
for (int i = n; i >= 1; i--) {
   for (int i = 1; j <= i; j++)  System.out.print("* ");
```

quick 2 knowledge

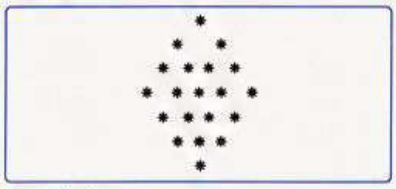
JAVA PATTERN NOTES

Java

All Important Patterns in Java

10. Diamond Pattern

Pattern:



Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= n - i; j++) System.out.print("*");
   for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
      System.out.println();
}
for (int i = n - 1; i >= 1; i--) {
   for (int j = 1; j <= n - i; j++) System.out.print("*");
   for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
      System.out.println();
}
```

11. Number Pyramid

Pattern:

```
        1
      2   2
    3   3   3
  4   4   4   4
5   5   5   5   5
```

Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++)
         System.out.print(i + " ");

      System.out.println();
}
```

12. Palindrome Pattern

Pattern:

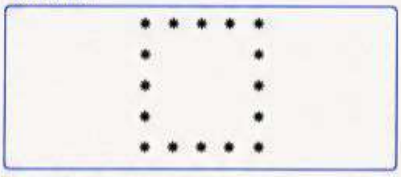
```
          1
        2 1 2
      3 2 1 2 3
    4 3 2 1 2 3 4
  5 4 3 2 1 2 3 4 5
```

Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   // increasing part
   for (int j = i; j >= 1; j--)  System.out.print(j + " ");
   // decreasing part
   for (int j = 2; j <= i; j++)  System.out.print(j + " ");
      System.out.println();
}
```

13. Hollow Square

Pattern:

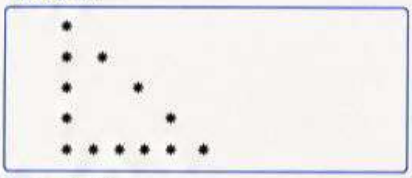


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= n; j++) {
      if (i == 1 || i == n || j == 1 || j == n)
         System.out.print("*");
      else
         System.out.print(" ");
   }
   System.out.println();
}
```

14. Hollow Right Triangle

Pattern:

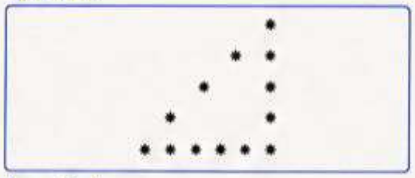


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++) {
      if (i == n || j == 1 || j == i)
         System.out.print("*");
      else
         System.out.print(" ");
   }
   System.out.println();
}
```

15. Hollow Left Triangle

Pattern:

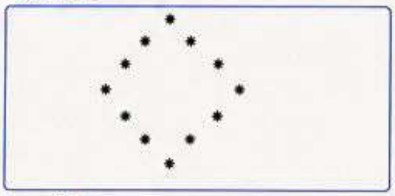


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= i; j++) {
      if (i == n || j == 1 || j == i)
         System.out.print("*");
      else
         System.out.print(" ");
   }
   System.out.println();
}
```

16. Hollow Diamond

Pattern:

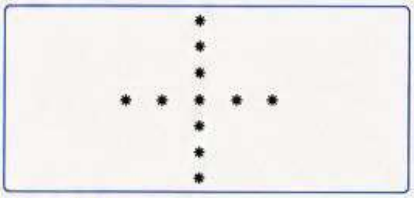


Java Code:

```
int n = 4;
// upper part
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= n - i; j++) System.out.print(" ");
   for (int j = 1; j <= 2  i - 1; j++) {
      if (j == 1 || j == 2 * i - 1 || i == n)
         System.out.print("*");
      else
         System.out.print(" ");
   }
   System.out.println();
// lower part
for (int i = n - 1; i >= 1; i--) {
   for (int j = 1; j <= n - i; j++) System.out.print(" ");
```

17. Plus Pattern

Pattern:



Java Code:

```
int n = 5;
int mid = n / 2;
for (int i = 0; i < n; i++) {
   for (int j = 0; j < n; j++) {
      if (i == mid || j == mid)
         System.out.print("*");
      else
         System.out.print(" ");
   }
```

18. X Pattern

Pattern:



Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= n; j++) {
      if (j == i || j == n - i + 1)
         System.out.print("*");
      else
         System.out.print(" ");
   }
   System.out.println();
```

quick 2 knowledge

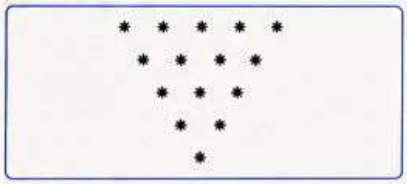
JAVA PATTERN NOTES

Java

All Important Patterns in Java

19. Inverted Right Triangle

Pattern:



Java Code:

```
int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```

20. Inverted Left Triangle

Pattern:

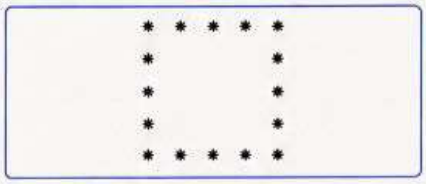


Java Code:

```
int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
    System.out.println();
}
```

21. Hollow Square

Pattern:

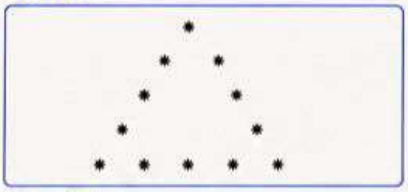


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (i == 1 || i == n || j == 1 || j == n)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
}
```

22. Hollow Triangle

Pattern:

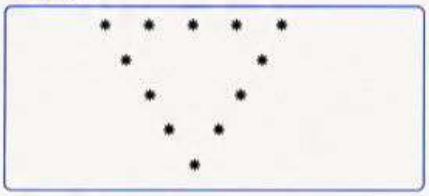


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= i; j++) {
        if (j == 1 || j == i || i == n)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
}
```

23. Hollow Inverted Triangle

Pattern:

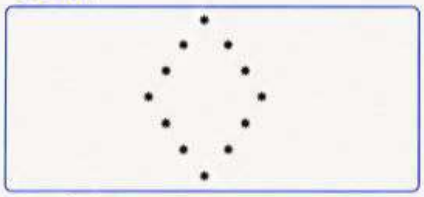


Java Code:

```
int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= i; j++) {
        if (j == 1 || j == i || i == n)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
}
```

24. Hollow Diamond

Pattern:

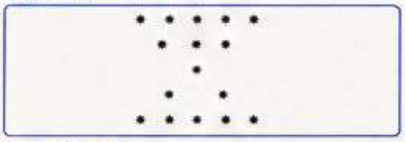


Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++){
        System.out.print("  ");
    for (int j = 1; j <= 2 * i - 1; j++) {
        if (j == 1 || j == 2 * i - 1 || i == n)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
} System.out.println();
// lower part
for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= 2 * i - 1; j++) {
        if (j == 1 || j == 2 * i - 1 || i == 1)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
} System.out.println();
```

25. Hourglass Pattern

Pattern:

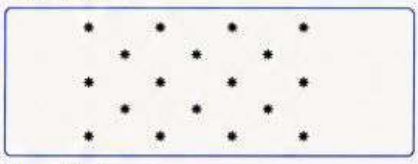


Java Code:

```
int n = 5;
// upper part
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= 2 * i - 1; j++)
        System.out.print("* ");
    System.out.println();
}
// lower part
for (int i = 2; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= 2 * i - 1; j++)
```

26. Zigzag Pattern

Pattern:

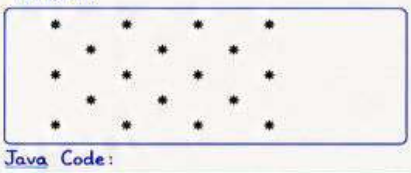


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if ((i + j) % 2 == 0)
            System.out.print("* ");
        else
            System.out.print("  ");
```

27. Wave Pattern

Pattern:



Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (j % 2 == 1) { // odd column
            if (i % 2 == 1)
                System.out.print("* ");
            else
                System.out.print("  ");
        } else { // even column
            if (i % 2 == 0)
                System.out.print("* ");
            else
                System.out.print("  ");
```

quick 2 knowledge

JAVA PATTERN NOTES

Java

All Important Patterns in Java

28. Checkerboard Pattern

Pattern:

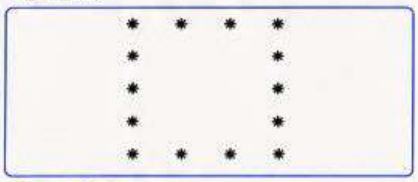


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (i + j) % 2 == 0)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
}
```

29. Box Pattern

Pattern:

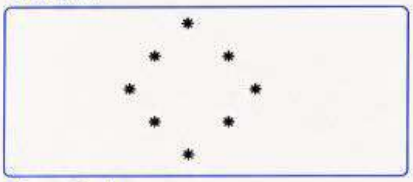


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (i == 1 || i == n ||
            j == 1 || j == n)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
}
```

30. Hollow Rhombus

Pattern:

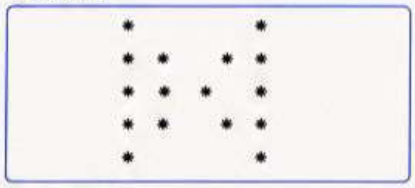


Java Code:

```
int n = 5;
int mid = n / 2 + 1;
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= n; j++) {
    if (i + j == mid + 1 || j - i == mid - 1)
        System.out.print("* ");
    else
        System.out.print("  ");
  }
  System.out.println();
}
```

31. Hollow Butterfly

Pattern:

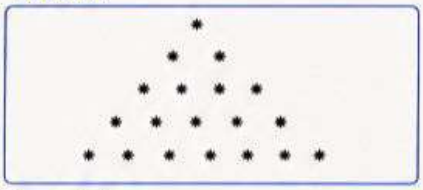


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= n; j++) {
    if (i == 1 || i == n ||j == 1 ||
        j == n || i == j || i + j == n + 1)
        System.out.print("* ");
    else
        System.out.print(" ");
  }
  System.out.println();
}
```

32. Tree Pattern

Pattern:

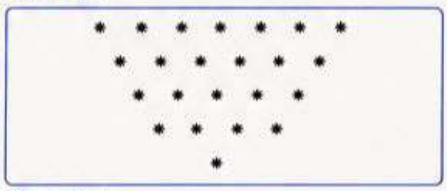


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print(" ");
    for (int j = 1; j <= 2  i  1; j++)
        System.out.print("* ");
    System.out.println();
}
```

33. Reverse Tree Pattern

Pattern:

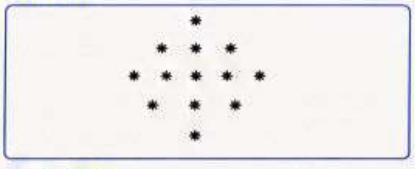


Java Code:

```
int n = 5;
for (int i = n; i >= 1; i--) {
  for (int j = 1; j <= n  i; j++)
      System.out.print(" ");
  for (int j = 1; j <= 2 * i - 1; j++)
      System.out.print("* ");
  System.out.println();
}
```

34. Diamond Pyramid

Pattern:

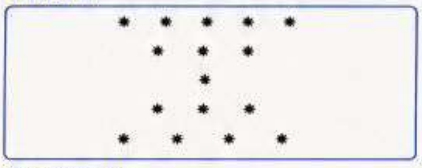


Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= 2 * i - 1; j++)
        System.out.print("* ");
    System.out.println();
}
// lower part
for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= 2 * i - 1; j++)
```

35. Rain Drop Pattern

Pattern:

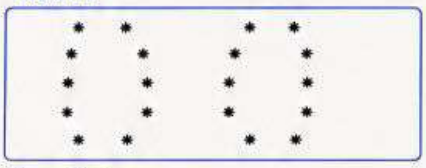


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= 2 * i - 1; j++)
        System.out.print("* ");
    System.out.println();
}
for (int i = n  1; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print("  ");
    for (int j = 1; j <= 2 * i  1; j++)
        System.out.print("* ");
```

36. Binoculars Pattern

Pattern:



Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (j <= i || j >= n - i + 1)
            System.out.print("* ");
        else
            System.out.print("  ");
    }
    System.out.println();
```

quick 2 knowledge

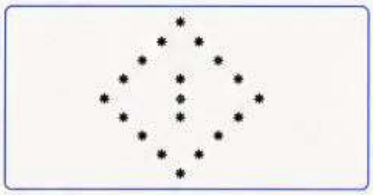
JAVA PATTERN NOTES

Java

All Important Patterns in Java

46. Diamond Inside Diamond

Pattern:



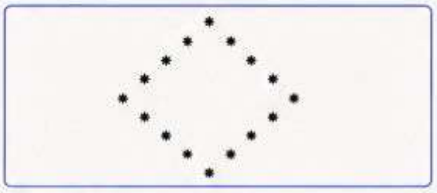
Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= n - i; j++) System.out.print(" ");
  for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
  for (int j = 1; j <= n - i; j++) System.out.print(" ");
     System.out.println();
}

// lower part
for (int i = n - 1; i >= 1; i--) {
  for (int j = 1; j <= n - i; j++) System.out.print(" ");
  for (int j = 1; j <= 2 * i - 1; j++) System.out.print("*");
  for (int j = 1; j <= n - i; j++) System.out.print(" ");
     System.out.println();
}
```

47. Hollow Diamond

Pattern:

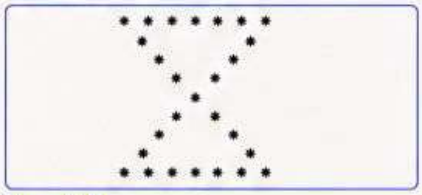


Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= n - i; j++)  System.out.print(" ");
   for (int j = 1; j <= 2 * i - 1; j++) {
       if (j == 1 || j == 2 * i - 1 || i == n)
           System.out.print("*");
       else
           System.out.print(" ");
   }
   for (int j = 1; j <= n - i; j++) System.out.print(" ");
      System.out.println();
}
// lower part
for (int i = n - 1; i >= 1; i--) {
   for (int j = 1; j <= n - i; j++) System.out.print(" ");
   for (int j = 1; j <= 2 * i - 1; j++) {
       if (j == 1 || j == 2 * i - 1 || i == 1)
           System.out.print("*");
       else
           System.out.print(" ");
   }
   for (int j = 1; j <= n - i; j++) System.out.print(" ");
}     System.out.println();
```

48. Sandglass Pattern

Pattern:

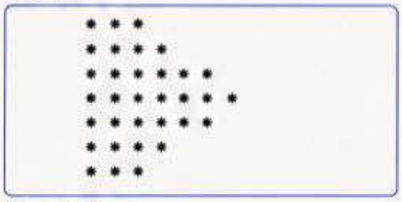


Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= i - 1; j++) System.out.print(" ");
  for (int j = 1; j <= 2 * (n - i) + 1; j++)
            System.out.print("*");
    System.out.println();
}
// lower part
for (int i = n - 1; i >= 1; i--) {
  for (int j = 1; j <= i - 1; j++) System.out.print(" ");
  for (int j = 1; j <= 2 * (n - i) + 1; j++)
            System.out.print("*");
    System.out.println();
}
```

49. Arrow Pattern

Pattern:

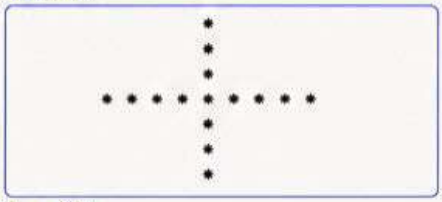


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= n - i; j++) System.out.print("*");
  for (int j = 1; j <= 2 * i - n - i; j++) System.out.print("*");
  for (int j = 1; j <= i; j++)  System.out.print("*");
     System.out.println();
}
```

50. Cross Pattern

Pattern:

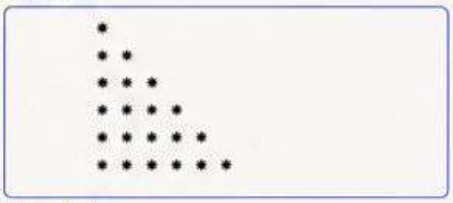


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
   for (int j = 1; j <= n; j++) {
       if (i == j || j == n - i + 1)
           System.out.print("*");
       else
           System.out.print(" ");
   }
   System.out.println();
```

51. Star Triangle

Pattern:



Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= i; j++) System.out.print("*");
    System.out.println();
}
```

52. Number Triangle

Pattern:

```
    1
  1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= i; j++) System.out.print(j + " ");
    System.out.println();
```

53. Palindrome Number Pattern

Pattern:

```
    1
  2 3 2
4 5 6 5 4
7 8 9 8 7
10 11 12 11 10
```

Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= 2 * i - 1; j++) System.out.print(j + " ");
  for (int j = 2 * i - 2; j >= i; j--) System.out.print(j + " ");
```

54. Triangle of Numbers

Pattern:

```
    1
  2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Java Code:

```
int n = 5, num = 1;
for (int i = 1; i <= n; i++) {
  for (int j = 1; j <= i; j++) System.out.print(num++ + " ");
```

quick 2 knowledge

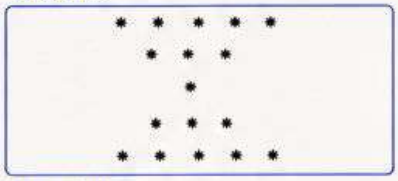
JAVA PATTERN NOTES

Java

All Important Patterns in Java

37. Hourglass

Pattern:

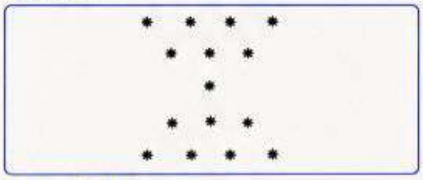


Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print(" ");
    for (int j = 1; j <= (2*i - 1); j++)
        System.out.print("*");
    System.out.println();
}
// lower part
for (int i = n-1; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print(" ");
    for (int j = 1; j <= (2*i - 1); j++)
        System.out.print("*");
    System.out.println();
}
```

38. Double Hourglass

Pattern:

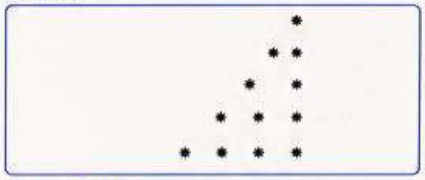


Java Code:

```
int n = 5;
// upper part
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print(" ");
    for (int j = 1; j <= (2*i - 1); j++)
        System.out.print("*");
    System.out.println();
}
// lower part
for (int i = 2; i <= n; i++) {
    for (int j = 1; j <= i - 1; j++)
        System.out.print(" ");
    for (int j = 1; j <= (2*(n-i+1)-1); j++)
        System.out.print("*");
    System.out.println();
}
```

39. Mirror Right Triangle

Pattern:

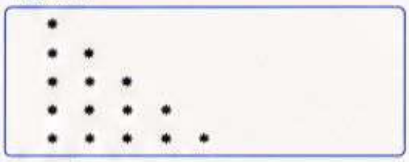


Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print(" ");
    for (int j = 1; j <= i; j++)
        System.out.print("*");
    System.out.println();
}
```

40. Mirror Left Triangle

Pattern:



Java Code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++)
        System.out.print("*");
    System.out.println();
}
```

41. Pascal's Triangle

Pattern:

```
        1
      1   1
    1   2   1
  1   3   3   1
1   4   6   4   1
```

Java Code:

```
int n = 5;
for (int i = 0; i < n; i++) {
    for (int j = 0; j <= i; j++) {
        int val = 1;
        for (int k = 0; k < j; k++)
            val = val * (i - k) / (k + 1);
        System.out.print(val + " ");
    }
    System.out.println();
}
```

42. Number Triangle

Pattern:

```
        1
      2   3
    4   5   6
  7   8   9   10
11  12  13  14  15
```

Java Code:

```
int n = 5, num = 1;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print(num + " ");
        num++;
    }
    System.out.println();
}
```

43. Spiral Pattern

Pattern:

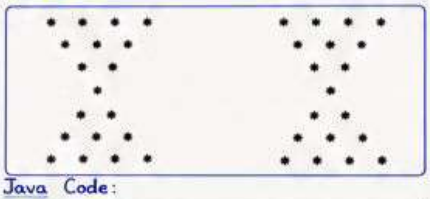
```
 1   2   3   4   5
16  17  18  19   6
15  24  25  20   7
14  23  22  21   8
13  12  11  10   9
```

Java Code:

```
int n = 5, num = 1;
int top = 0, bottom = n-1, left = 0, right = n-1;
while (top <= bottom && left <= right) {
    for (int j = left; j <= right; j++)
        System.out.print(num++ + " ");
    top++;
    for (int i = top; i <= bottom; i++)
        System.out.print(num++ + " ");
    right--;
    if (top <= bottom) {
        for (int j = right; j >= left; j--)
            System.out.print(num++ + " ");
        bottom--;
    }
    if (left <= right) {
        for (int i = bottom; i >= top; i--)
```

44. Butterfly Pattern

Pattern:

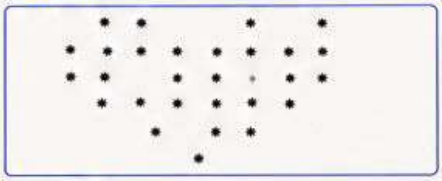


Java Code:

```
int n = 4;
// upper part
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i + 1; j++)
        System.out.print("* ");
    for (int j = 1; j <= 2 * (i - 1); j++)
        System.out.print("  ");
    for (int j = 1; j <= n - i + 1; j++)
        System.out.print("* ");
    System.out.println();
}
// lower part
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= n - i + 1; j++)
        System.out.print("* ");
    for (int j = 1; j <= 2 * (i - 1); j++)
        System.out.print("  ");
```

45. Heart Pattern

Pattern:



Java Code:

```
int n = 4;
// upper part
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
    for (int j = 1; j <= 2 * (n - i); j++)
        System.out.print("  ");
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
    System.out.println();
}
// lower part
for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
```